



| Number | Question                                                                                             | Answer                                                                                                                                              |
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| 1      | What is the relationship between the autocorrelation function and the mean square value of a signal? | The autocorrelation function at zero lag, $R_{xx}(0)$ , equals the mean square value: $R_{xx}(0) = E[x^2(t)]$ .                                     |
| 2      | What is the definition of the Z-transform for a discrete-time signal $x[n]$ ?                        | $X(z) = \sum_{n=-\infty}^{\infty} x[n] * z^{-n}$ .                                                                                                  |
| 3      | What is Numerical Differentiation?                                                                   | It is a method that approximates the derivative of a function using discrete data points.                                                           |
| 4      | What is Bayes' formula or Bayes' theorem?                                                            | $P(A B) = [P(B A) * P(A)] / P(B)$ .                                                                                                                 |
| 5      | What is the relationship between the poles of a system and its eigenvalues?                          | For a system described in state-space form, the poles are the eigenvalues of the system matrix.                                                     |
| 6      | What is the purpose of a root locus plot?                                                            | To determine the stability of a closed-loop control system for different values of a gain parameter.                                                |
| 7      | What is the Second Moment of Area?                                                                   | It is a geometrical property of a cross-section, also known as the Area Moment of Inertia, which predicts the resistance to bending and deflection. |
| 8      | What is the effect of a system having a transfer function with complex conjugate zeros?              | It can cause a dip or notch in the frequency response magnitude at a specific frequency.                                                            |
| 9      | What is the effect of a decrease in step size in numerical differentiation?                          | It generally decreases truncation error but may increase round-off error due to more computational steps.                                           |
| 10     | In hypothesis testing, what is the level of significance ( $\alpha$ )?                               | It is the probability of rejecting the null hypothesis when it is actually true (Type I error).                                                     |
| 11     | How are the zeros and the number of poles of a transfer function determined?                         | The zeros are the roots of the numerator polynomial. The number of poles is equal to                                                                |

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|    |                                                                                                     | the order of the denominator polynomial.                                                                              |
| 12 | What is an Integrating Factor used for?                                                             | It is used to solve First Order Linear Differential Equations.                                                        |
| 13 | What is a Jacobian Matrix?                                                                          | It is a matrix containing all the first-order partial derivatives of a vector-valued function.                        |
| 14 | What are the roots of the numerator of the closed-loop transfer function called?                    | They are called the zeros of the closed-loop system.                                                                  |
| 15 | Which statement is TRUE about conditional probability?                                              | $P(A B) = P(A \cap B) / P(B)$ , provided $P(B) > 0$ .                                                                 |
| 16 | Which of the following is TRUE for independent events A and B?                                      | $P(A \cap B) = P(A) \times P(B)$ .                                                                                    |
| 17 | Which of the following is an example of the complement rule?                                        | $P(A') = 1 - P(A)$ .                                                                                                  |
| 18 | What is the formula for the probability of event A or B occurring?                                  | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .                                                                           |
| 19 | What type of equation is $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$ ?          | Laplace's Equation.                                                                                                   |
| 20 | What statistical method is used for comparing the means of two groups?                              | The t-test.                                                                                                           |
| 21 | What is the unit impulse response of a system?                                                      | It is the output $y(t)$ of a system when the input is a unit impulse $\delta(t)$ and all initial conditions are zero. |
| 22 | If events A and B are mutually exclusive, which of the following is TRUE?                           | $P(A \cap B) = 0$ .                                                                                                   |
| 23 | In hypothesis testing, what happens when the probability of a Type I error is decreased?            | It generally increases the probability of a Type II error.                                                            |
| 24 | What test statistic is used in Analysis of Variance (ANOVA)?                                        | The F-statistic.                                                                                                      |
| 25 | What type of equation is $\partial u / \partial t - k \partial^2 u / \partial x^2 = 0$ ?            | The Diffusion Equation or Heat Equation.                                                                              |
| 26 | What allows for a complex plane representation of a digital signal or system using poles and zeros? | The Z-transform.                                                                                                      |
| 27 | What is the derivative of $f(x) = x^x$ ?                                                            | $f'(x) = x^x (\ln(x) + 1)$ .                                                                                          |
| 28 | What is the Z-transform of the sequence $a^n * u[n]$ ?                                              | $X(z) = 1 / (1 - a z^{-1})$ , for $ z  >  a $ .                                                                       |
| 29 | What is the relationship between cross-correlation and convolution?                                 | Cross-correlation is similar to convolution without the time-reversal of one of the signals.                          |

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| 30 | In regression analysis, what does the least squares method minimize? | The sum of the squared residuals (errors). |
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| 30 | In regression analysis, what does the least squares method minimize?                                      | The sum of the squared residuals (errors).                                                                                                                            |
| 31 | What is the geometric representation of the magnitude of the cross product of two vectors?                | It represents the area of the parallelogram formed by the two vectors.                                                                                                |
| 32 | Which test is used to compare two population variances?                                                   | The F-test.                                                                                                                                                           |
| 33 | What is the probability that at least one of two independent events occurs?                               | $P(A \cup B) = P(A) + P(B) - P(A)P(B)$ .                                                                                                                              |
| 34 | In continuous-time systems (s-plane), when is a system stable?                                            | If all poles lie in the left half of the s-plane.                                                                                                                     |
| 35 | What are forward, backward, and central differences used for?                                             | To approximate the derivatives of functions from discrete data.                                                                                                       |
| 36 | What property of a PDE is demonstrated if a linear combination of solutions is also a solution?           | The Superposition Principle.                                                                                                                                          |
| 37 | What type of curve does the Trapezoidal rule use to approximate an integral?                              | A straight line segment (linear approximation).                                                                                                                       |
| 38 | What is the convolution of a continuous-time signal $x(t)$ with a shifted impulse $\delta(t - T)$ ?       | $x(t - T)$ .                                                                                                                                                          |
| 39 | What is Laplace's equation typically used to model?                                                       | Steady-state potential fields, such as in electrostatics or fluid flow.                                                                                               |
| 40 | In which direction from a point does a scalar field increase most rapidly?                                | The direction of the gradient vector at that point.                                                                                                                   |
| 41 | What type of approximation does the Trapezoidal rule use?                                                 | Linear approximation.                                                                                                                                                 |
| 42 | What is non-response bias?                                                                                | It occurs when survey participants are unable or unwilling to respond, leading to a biased sample.                                                                    |
| 43 | What is the mathematical operation for the convolution of two continuous-time signals $h(t)$ and $g(t)$ ? | $h(t) * g(t)$ .                                                                                                                                                       |
| 44 | What is the Divergence Theorem?                                                                           | $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$ .<br>It relates the flux through a closed surface to the divergence within the volume. |
| 45 | What is the directional derivative of $\Phi = xyz$ in the direction of $(1, 0, 0)$ ?                      | The directional derivative is $(yz)$ at the point, which is the result of the dot product between the gradient $(yz, xz,$                                             |

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|    |                                                                                                                    | xy) and the unit vector (1, 0, 0).                                                  |
| 46 | What is the formula for the Z-transform of a discrete-time signal $x[n]$ ?                                         | $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$ .                                    |
| 47 | In a discrete-time system (z-plane), where must the poles be located for stability?                                | Inside the unit circle.                                                             |
| 48 | What property of the Laplace Transform is expressed by $L\{af(t) + bg(t)\} = aF(s) + bG(s)$ ?                      | The Linearity Property.                                                             |
| 49 | In numerical integration, how can accuracy be improved?                                                            | By increasing the number of steps (decreasing the step size).                       |
| 50 | Which distribution models the number of successes in a fixed number of independent Bernoulli trials?               | The Binomial distribution.                                                          |
| 51 | What is the curl of the vector field $F = (x, y, z)$ ?                                                             | (0, 0, 0).                                                                          |
| 52 | What is the divergence of the vector field $F = (x, y, z)$ ?                                                       | 3.                                                                                  |
| 53 | What is a solution of a PDE that contains arbitrary functions called?                                              | The General Solution.                                                               |
| 54 | What is the relationship between convolution and cross correlation?                                                | Convolution involves time-reversal of one signal, while cross-correlation does not. |
| 55 | The birthday paradox shows that in a group of 23 people, what is the approximate probability of a shared birthday? | About 50%.                                                                          |
| 56 | What is the transient response of a control system?                                                                | The part of the response that exists before it reaches a steady state.              |
| 57 | What is the effect of having poles in the left half of the s-plane on the time-domain response?                    | It causes a decaying exponential response, leading to stability.                    |
| 58 | In hypothesis testing, what is the conclusion if the p-value is greater than alpha ( $\alpha$ )?                   | Fail to reject the null hypothesis.                                                 |
| 59 | What is a major source of error in representing signals numerically?                                               | Truncation error.                                                                   |
| 60 | What is the autocorrelation function of $x(t) = \sin(2\pi ft)$ ?                                                   | $R_{xx}(\tau) = (1/2) \cos(2\pi f\tau)$ .                                           |
| 61 | What happens to the width of a confidence interval when the sample size increases?                                 | It becomes narrower.                                                                |
| 62 | What theorem states that at least $(1 - 1/k^2)$ of data lies within k standard deviations of the mean?             | Chebyshev's Inequality.                                                             |
| 63 | What is a defining requirement for a Partial Differential Equation (PDE)?                                          | It must involve two or more independent variables.                                  |

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| 64 | Which of the following is TRUE regarding Green's and Stokes' Theorems?                                      | Green's Theorem is a special case of Stokes' Theorem in two dimensions.                                    |
| 65 | What happens to the probability of a Type II error when the probability of a Type I error is made very low? | It increases.                                                                                              |
| 66 | What is the Second Moment of Area also known as?                                                            | Area Moment of Inertia.                                                                                    |
| 67 | If the significance level ( $\alpha$ ) is 0.05, what does it represent?                                     | The probability of rejecting the null hypothesis ( $H_0$ ) when it is actually true (Type I error).        |
| 68 | What does the directional derivative in a vector field represent?                                           | The rate of change of the field quantity in a specified direction.                                         |
| 69 | A Fourier series with only sine terms represents what kind of function?                                     | An odd function.                                                                                           |
| 70 | What does the Fourier Series represent?                                                                     | A periodic continuous-time signal as a sum of sinusoidal harmonics.                                        |
| 71 | What is the effect of poles in the left-half of the s-plane on the transient response?                      | A decaying response.                                                                                       |
| 72 | If the standard deviation of a dataset is doubled, what happens to the variance?                            | The variance becomes four times larger (it quadruples).                                                    |
| 73 | In hypothesis testing, the significance level ( $\beta$ ) is the probability of what?                       | Beta ( $\beta$ ) is the probability of making a Type II error (failing to reject a false null hypothesis). |
| 74 | Find the divergence of the vector field $F = (x^2, y^2, z^2)$ .                                             | $2x + 2y + 2z$ .                                                                                           |
| 75 | What is the use of poles and zeros in a system analysis?                                                    | They are used to determine stability, and analyze transient and frequency response (all of the above).     |
| 76 | What is the operation $x[-n]$ called?                                                                       | Time reversal.                                                                                             |
| 77 | What is the effect of moving a pole further to the left in the s-plane?                                     | The system becomes more stable, with a faster transient decay.                                             |
| 78 | What does the Fourier Transform convert?                                                                    | A signal from the Time Domain to the Frequency Domain.                                                     |
| 79 | What determines the transient response of a control system?                                                 | The poles of the system's transfer function.                                                               |





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| 96  | What theorem states that the sampling distribution of the sample mean approximates a normal distribution for a large sample size? | The Central Limit Theorem (CLT).                                                                                                    |
| 97  | What is another name for Stokes' Theorem?                                                                                         | The Curl Theorem.                                                                                                                   |
| 98  | What happens to the confidence interval width as the sample size increases?                                                       | It decreases (becomes narrower).                                                                                                    |
| 99  | What is the formula for Stokes' Theorem?                                                                                          | $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ .                                     |
| 100 | What is the formula for the directional derivative of a scalar field $V$ in the direction of unit vector $\hat{a}$ ?              | $\nabla V \cdot \hat{a}$ .                                                                                                          |
| 101 | Which theorem relates a line integral around a closed curve to a double integral over the enclosed region?                        | Green's Theorem.                                                                                                                    |
| 102 | In an ANOVA table, how is the F-ratio calculated?                                                                                 | Mean Square Between Groups (Treatment) divided by Mean Square Within Groups (Error).                                                |
| 103 | What is the general form of the solution to a linear differential equation?                                                       | A combination of exponential and constant functions, or more generally, functions whose derivatives are proportional to themselves. |
| 104 | A system with two complex conjugate poles is typically:                                                                           | Underdamped.                                                                                                                        |
| 105 | The poles of a system determine the:                                                                                              | Transient response.                                                                                                                 |
| 106 | What is the primary application of the Divergence Theorem?                                                                        | Calculating the flux of a vector field through a closed surface.                                                                    |
| 107 | What is the formula for Green's Theorem in the plane?                                                                             | $\oint_C (P dx + Q dy) = \iint_D (\partial Q / \partial x - \partial P / \partial y) dA$ .                                          |
| 108 | What is the curl of the vector field $\mathbf{F} = (y^2, x^2, z^2)$ ?                                                             | $(0, 0, 2x - 2y)$ .                                                                                                                 |
| 109 | What does it mean for two vectors to be orthogonal?                                                                               | Their dot product is zero.                                                                                                          |
| 110 | If the variance doubles, what happens to the standard deviation?                                                                  | It increases by a factor of $\sqrt{2}$ .                                                                                            |
| 111 | When is a discrete-time system causal?                                                                                            | If its output at any time depends only on the present and past values of the input.                                                 |
| 112 | What defines a linear differential equation?                                                                                      | The dependent variable and its derivatives appear only to the first power and are not multiplied together.                          |
| 113 | When is the directional derivative of a scalar field maximum?                                                                     | When the direction is parallel to the gradient vector.                                                                              |

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| 114 | Which fundamental law is derived using the Divergence Theorem in electromagnetics? | Gauss's Law.                                                                                                               |
| 115 | Which of the following is TRUE about Gauss's Law and the Divergence Theorem?       | Gauss's Law for magnetism is a special case of the Divergence Theorem.                                                     |
| 116 | As the damping ratio increases, what happens to the frequency response?            | It becomes less oscillatory (the resonance peak decreases).                                                                |
| 117 | What is the equation for the gradient of a scalar field $\phi$ ?                   | $\nabla\phi = (\partial\phi/\partial x, \partial\phi/\partial y, \partial\phi/\partial z)$ .                               |
| 118 | What is the variance of the sample mean?                                           | $\sigma^2 / n$ , where $\sigma^2$ is the population variance and $n$ is the sample size.                                   |
| 119 | What is the divergence of the vector field $F = (0, 0, 0)$ ?                       | 0.                                                                                                                         |
| 120 | As the damping ratio increases, what happens to the peak overshoot?                | It decreases.                                                                                                              |
| 121 | What is the significance of the Divergence Theorem?                                | It relates the flux of a vector field through a closed surface to the divergence of that field within the enclosed volume. |
| 122 | In a one-way ANOVA, what does it suggest if the F-statistic is smaller than 1?     | There is no evidence that the group means are different; they are similar.                                                 |
| 123 | As the damping ratio decreases, what happens to the settling time?                 | It increases.                                                                                                              |
| 124 | What is the formula for the median of a dataset (1,1,1,2,3)?                       | The median is 1 (the middle value when sorted).                                                                            |
| 125 | Which of the following represents the mean of a binomial distribution?             | $\mu = n * p$ .                                                                                                            |
| 126 | What is the gradient of $\Phi = x^2 + y^2 + z^2$ at the point (1,1,1)?             | (2, 2, 2).                                                                                                                 |
| 127 | When is a Type II error most likely to occur?                                      | When the sample size is too small to detect an actual effect.                                                              |
| 128 | A linear system with distinct real eigenvalues has a solution composed of what?    | Exponential functions.                                                                                                     |
| 129 | What is the direction of the gradient vector relative to a level surface?          | It is perpendicular (normal) to the level surface.                                                                         |
| 130 | What is a Jacobian matrix?                                                         | A matrix of first-order partial derivatives of a vector-valued function.                                                   |

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| 131 | As the confidence level increases (e.g., from 95% to 99%), what happens to the confidence interval?   | It becomes wider.                                                                                                                       |
| 132 | What is the formula for the probability of the union of two events?                                   | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .                                                                                             |
| 133 | Stokes' Theorem is dependent on what?                                                                 | The orientation of the surface and its boundary curve.                                                                                  |
| 134 | What is the general formula for the second moment of inertia with respect to a distance $r$ ?         | $\int r^2 dA$ .                                                                                                                         |
| 135 | What pole-zero configuration gives a root locus with an asymptote angle of $90^\circ$ ?               | A system with 2 poles and no zeros.                                                                                                     |
| 136 | What is the Maclaurin series for $1/(1-x)$ , and what is its interval of convergence?                 | $\sum$ (from $n=0$ to $\infty$ ) $x^n$ , for $ x  < 1$ .                                                                                |
| 137 | How is conditional probability represented?                                                           | $P(A B) = P(A \cap B) / P(B)$ .                                                                                                         |
| 138 | How are the poles of a system determined?                                                             | They are the roots of the denominator polynomial of the transfer function.                                                              |
| 139 | In a discrete uniform distribution with $n$ outcomes, what is the probability of each outcome?        | $1/n$ .                                                                                                                                 |
| 140 | What is the transfer function $H(s)$ defined as?                                                      | The ratio of the Laplace transform of the output to the Laplace transform of the input.                                                 |
| 141 | What is the closed-loop transfer function for a system with forward gain $G(s)$ and feedback $H(s)$ ? | $G(s) / [1 + G(s)H(s)]$ .                                                                                                               |
| 142 | A PDE is linear if:                                                                                   | The dependent variable and its derivatives appear only to the first power and are not multiplied together.                              |
| 143 | What is the Shannon-Hartley theorem?                                                                  | It gives the maximum theoretical rate for error-free data transmission over a channel with a given bandwidth and signal-to-noise ratio. |
| 144 | What type of curve does Simpson's 1/3 rule use?                                                       | A parabolic arc.                                                                                                                        |